

1. Explain Types of Comments in Python

Comments are used to make code more readable and are ignored by the Python interpreter.

a) Single-Line Comment

A single-line comment starts with #.

```
# This is a single-line comment
print("Hello World")
```

b) Multi-Line Comment

Python does not have a dedicated multi-line comment syntax. Triple quotes (''' or ''') are commonly used for multi-line comments or docstrings.

```
"""
This is a
multi-line comment
"""
print("Hello World")
```

2. What are the 2 Ways to Execute a Python Program?

a) Interactive Mode

- Execute Python statements one by one.
- Open the Python interpreter and type commands directly.

Example:

```
>>> print("Hello")
Hello
```

b) Script Mode

- Write Python code in a .py file and execute the file.
- Suitable for larger programs.

Example:

```
# hello.py
print("Hello World")
```

Run using:

```
python hello.py
```

3. What is a Multi-Line Statement?

A multi-line statement is a statement that spans more than one line for better readability.

Using Backslash (\)

```
total = 10 + 20 + 30 + \
        40 + 50
print(total)
```

Using Parentheses

```
total = (10 + 20 + 30 +
        40 + 50)
print(total)
```

Output:

```
150
```

A multi-line statement helps make long expressions easier to read and maintain.